Production Information System

CHAPTER 3: Data Modeling

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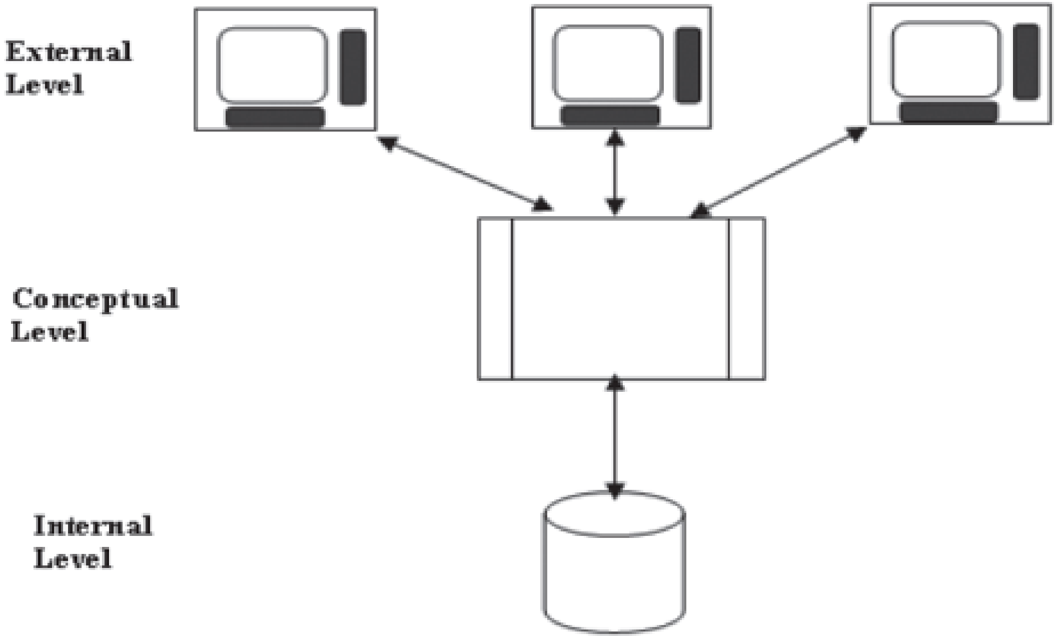
Outline

1. Introduction
2. Entity-Relationship (E-R) Modeling
3. Normalization
4. Case Study in Data Modeling

1. Introduction

In 1975, a general architecture for data representation was developed by the Standards Planning and Requirements Committee (SPARC) of the American National Standards Institute (ANSI).

The ANSI/SPARC architecture is a three-level architecture based on three views of the data in the database:



1. Introduction

- The external level Addresses the way in which different users view the database
- Includes those entities and attributes that the user sees and interacts with
- The actual implementation involves user interfaces that are used to interact with the database
- These interfaces are typically designed as “forms” and “reports”
- The user “view” can also include representations of the data that are not stored in the database

1. Introduction

- The conceptual level is a holistic view of the database at the level that defines entities, their attributes, and their relationships
- Describes what data are stored in the database, but not how they are stored
- The conceptual level is a logical description of the database without saying anything about its implementation
  - Therefore, it is independent of any specific hardware or software platforms
- The conceptual level is the foundation for the external level design

1. Introduction

- The internal level addresses the data structures and the file organization that are used to store data within the computer
- Whereas the conceptual level defines what data are stored, the internal level defines how data are stored
- This includes:
  - Data compression
  - Data encryption
  - Use of indexes
  - Hashing schemes
  - Other details
- The internal level is dependent on the operating system and physical components of the computer system

1. Introduction

- In this chapter, we focus on the conceptual, or logical, design of the database:
- We refer to this as the design of a data model
- The data model provides a representation of:
  - The entities in the enterprise
  - The attributes of those entities
  - The relationships that exist among entities
- Several related modeling formalisms are used to develop relational data models
- We describe Chen's entity-relationship (E-R) modeling technique as the foundation of principles for data modeling

2. Entity-Relationship (E-R) Modeling

- The purpose of E-R modeling is to design a conceptual schema of entities and their relationships in order to:
- Facilitate the process of communication among analysts, managers, and users of data
- Design a common data model that will accommodate the different needs of individuals and organizations within the enterprise
- Establish a logical model that can be implemented in a database management system (DBMS)

2. Entity-Relationship (E-R) Modeling

- The E-R model is a conceptual model and does not depend on any particular hardware or software
- E-R modeling is an interactive process that is carried on between the designer of the database application and the individuals in the organization that is the subject of the data model

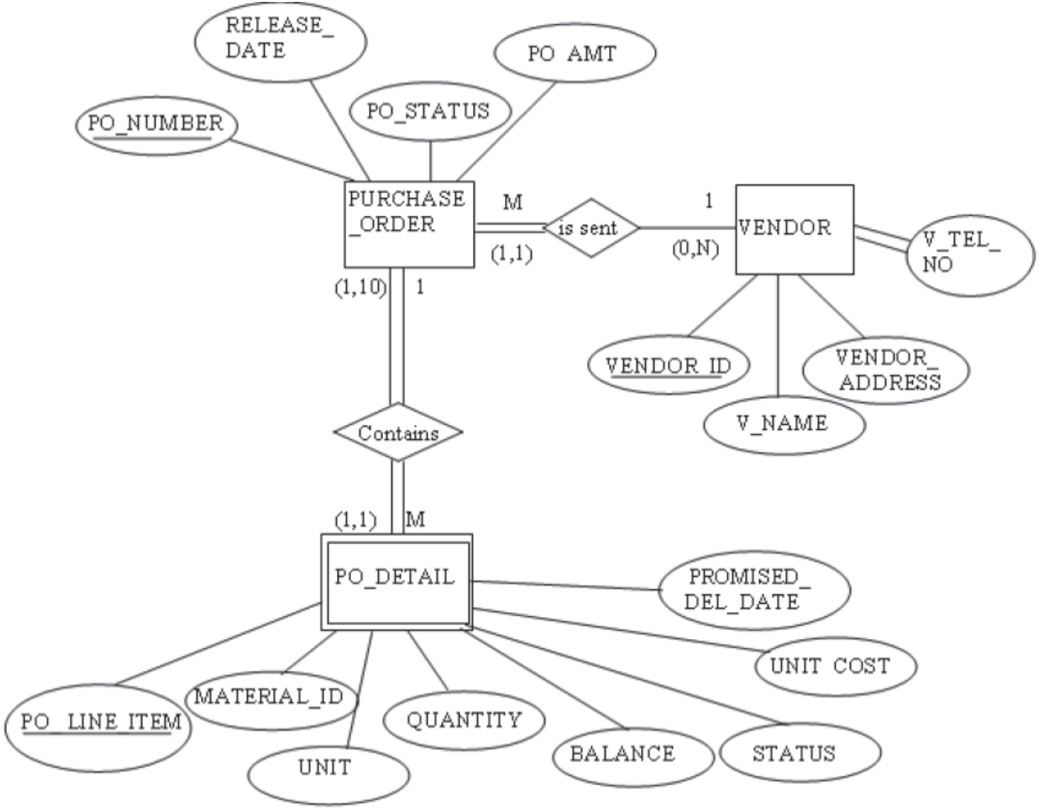
2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- The E-R modeling approach focuses on three aspects of data:
  - Entity sets
  - Attributes
  - Relationships between entity sets
- The E-R diagram uses a box to represent an entity set
- In the E-R modeling formalism:
  - The term “entity” describes what is referred to as “entity set”
  - The term “entity instance” describes what we refer to as an “entity” or “record”

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives



Entity Relationship (E-R) Diagram

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- E-R diagrams distinguish between weak and strong entities
  - An entity is weak if its existence is dependent on the existence of another entity
    - Shown in an E-R diagram by using a double-sided rectangle
    - Example: PO_DETAIL is dependent on the existence of PURCHASE_ORDER
    - Sometimes referred to as a child in a parent-child relationship
    - The primary key of the weak (child) entity is at least in part derived from the parent entity
  - A strong entity is not existence dependent on some other entity
    - Represented by a single-sided rectangle

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- In the E-R modeling formalism, attributes are represented by ovals with a line joining them to the entity
- Primary keys are indicated by underlining the attribute name
- Examples:
  - PO_NUMBER is the primary key attribute of PURCHASE_ORDER
  - VENDOR_ID is the primary key for VENDOR

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- In E-R models, Foreign keys are not shown
  - VENDOR_ID is the foreign key that links PURCHASE_ORDER to the entity VENDOR
- When tables are implemented from an E-R model, a table must contain, as a foreign key, the primary key of any table to which it is related.
  - Foreign keys will be inherited when tables are created from the E-R model
  - Therefore, it is unnecessary to show the attribute VENDOR_ID with the entity PURCHASE_ORDER. VENDOR_ID will be inherited by PURCHASE_ORDER when the tables are created

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- In the conceptual modeling stage, attributes are sometimes introduced without consideration as to how they would be implemented in the database tables
  - An example of this is introducing composite attributes into the model
- A composite attribute is an attribute that can be further broken down into more specific attributes
  - Example: VENDOR_ADDRESS can be subdivided into:
    - Street
    - City
    - State
    - Zip code

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- An attribute may be described as being either single valued or multivalued:
  - A single-valued Attribute will have only one entry in each instance of that attribute
    - Example: PO_NUMBER - only one purchase order number per entry
  - Other attributes may have more than one value (multivalued)
    - Example: Consider the attribute V_TEL_NO of the entity VENDOR. The vendor may have more than one primary telephone number that should be kept track of in the database.
    - At the early design stage, indicate with double lines connecting the attribute to the entity

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- The relationship is a description of the association between entities.
- Relationships have three components:
  - Connecting arcs
    - connect entities that are related to each other
    - VENDOR is related to PURCHASE_ORDER and PURCHASE_ORDER is related to PO_DETAIL
  - Diamonds
    - contain verb or verb phrase expressing the general relationship
    - VENDOR is sent PURCHASE_ORDER”
    - “PURCHASE_ORDER contains PO_DETAIL”
  - Cardinality
    - expresses the number of entity occurrences associated with one occurrence of the related entity.
    - Cardinality is indicated by numbers or symbols written at the ends of the relationship arc
  - Three common cardinalities on E-R diagrams:
    - One-to-many (1:M)
    - One-to-one (1:1)
    - Many-to-many (M:N)

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- The cardinality 0 : M means that each instance of one entity is associated with zero, one, or many instances of the entity on the M side of the relationship
- Examples:
  - “One vendor is sent 0, 1 or many purchase orders” (0:M)
  - “One purchase order contains 1 or many purchase order details” (1:M)

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- A one-to-one relationship often indicates that the two entities involved can be combined into a single entity.
- Why?
- A record in one table will be related to one and only one record in another table to which it has a one-to-one relationship. Therefore, no increase in the number of rows will occur if both entities are combined in one table.

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- A useful graphical tool that assists in visualizing the cardinality of a relationship is called a semantic net diagram
- Examples of the instances of two entities are connected by relationship lines

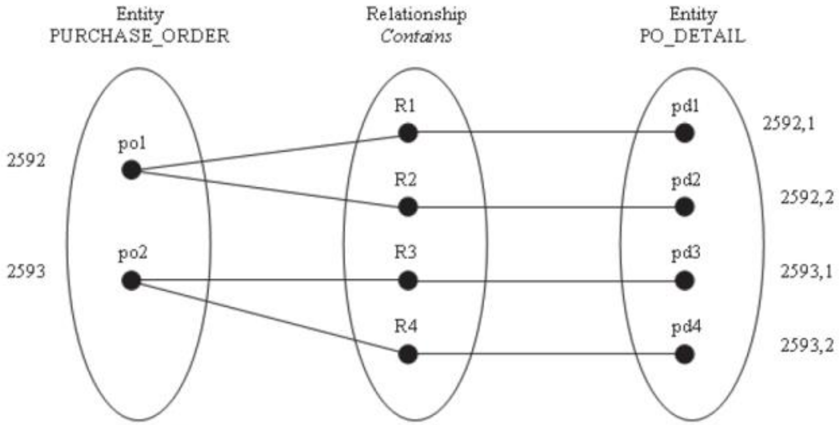


Figure 3.3 A 1:M relationship in a semantic net diagram.

Entity Set: VENDOR

VENDOR_ID	V_NAME	V_STREET	V_CITY	V_STATE	V_ZIP
V110	Jersey Vegetable Co.	2 Main St.	Patterson	NJ	07055
V25	General Provisions	125 Common St.	Boise	ID	44830
V250	Spices Unlimited	25 Salty Lane	East Hampton	NY	10027
V75	Pasta Supply, Inc.	34 Henry St.	Philadelphia	PA	09098

Entity Set: PURCHASE_ORDER

PO_NUMBER	RELEASE_DATE	PO_STATUS	PO_AMT	VENDOR_ID
2591	2/10/06	CLOSED	$4,300.00	V110
2592	2/10/06	OPEN	$505.50	V25
2593	2/11/06	OPEN	$4,000.00	V110
2594	2/12/06	OPEN	$3,280.00	V250
2595	2/15/06	OPEN	$500.00	V250
2596		HOLD	$1,000.00	V75

Entity Set: PO_DETAIL

PO_NUMBER	PO_LINE_IT	MATERIAL_ID	UNITS	QUANTITY	BALANCE	PROMISED_DEL_	UNIT_COST	STATUS
2591	1	RM201	LB	1000.0	0	2/20/06	$2.00	CLOSED
2591	2	RM202	LB	1000.0	0	2/20/06	$2.00	CLOSED
2591	3	RM205	LB	300.0	0	2/20/06	$1.00	CLOSED
2592	1	RM805	GAL	800.5	0	2/25/06	$0.50	CLOSED
2592	2	RM810	GAL	210.5	210	3/10/06	$0.50	OPEN
2594	1	RM310	LB	4000.0	4000	3/12/06	$0.50	OPEN
2594	2	RM311	LB	2000.0	2000	3/12/06	$0.25	OPEN
2594	3	RM318	LB	2000.0	2000	3/12/06	$0.25	OPEN
2594	4	RM340	LB	560.0	560	3/20/06	$0.50	OPEN
2593	1	RM210	LB	1000.0	500	2/25/06	$2.00	OPEN
2593	2	RM211	LB	2000.0	2000	3/10/06	$1.00	OPEN
2595	1	RM305	LB	400.0	400	2/27/06	$0.50	OPEN
2595	2	RM308	LB	1200.0	1200	2/27/06	$0.25	OPEN
2596	1	RM502	LB	5000.0	5000		$0.20	OPEN

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

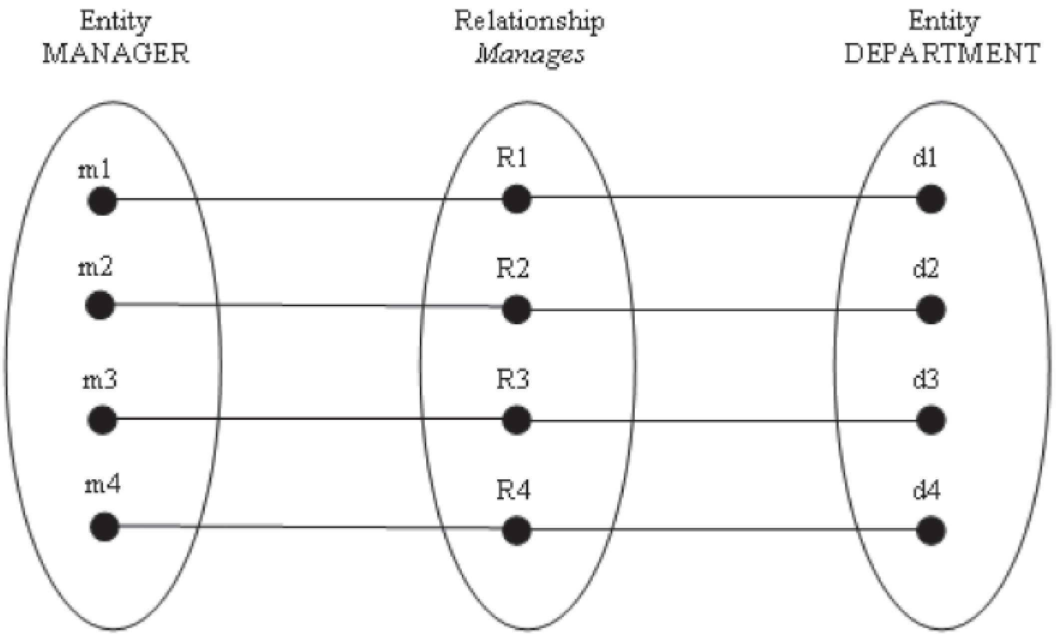


Figure 3.4 A 1:1 relationship in a semantic net diagram.

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

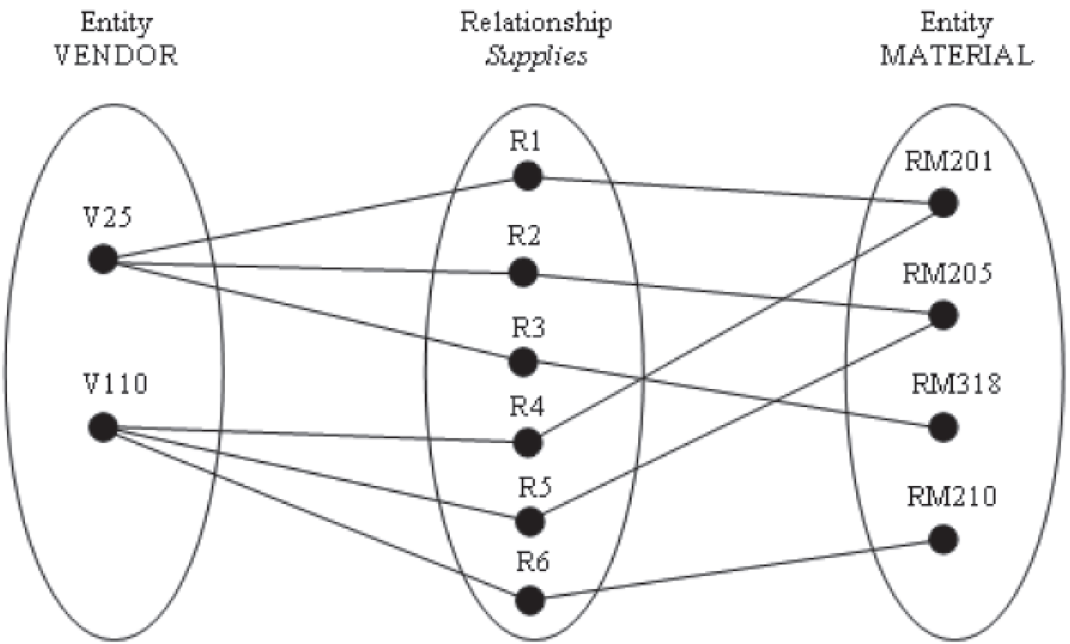


Figure 3.5 M:N Relationship in a semantic net diagram.

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

- It is often desirable to add cardinality limits to the E-R diagram to be more specific about the domain of the cardinality relationship
  – Cardinality limits are shown in E-R diagram (S.11) as numbers in parentheses
- Examples:
  – In the relationship between PURCHASE_ORDER and PO_DETAIL, the cardinality limits for PO_DETAIL are (1,1), which means that each PO_DETAIL will be associated with one purchase order
  – The cardinality limits on PURCHASE_ORDER is (1,10), which means that there will be a minimum of 1 and a maximum of 10 line items associated with one purchase order
- Cardinality limits are imposed based on the way in which the business is operated. Therefore, they are said to reflect the business rules of the enterprise.

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives

Business rules affect the degree of an entity's participation in a relationship.

There are two categories of entity participation:

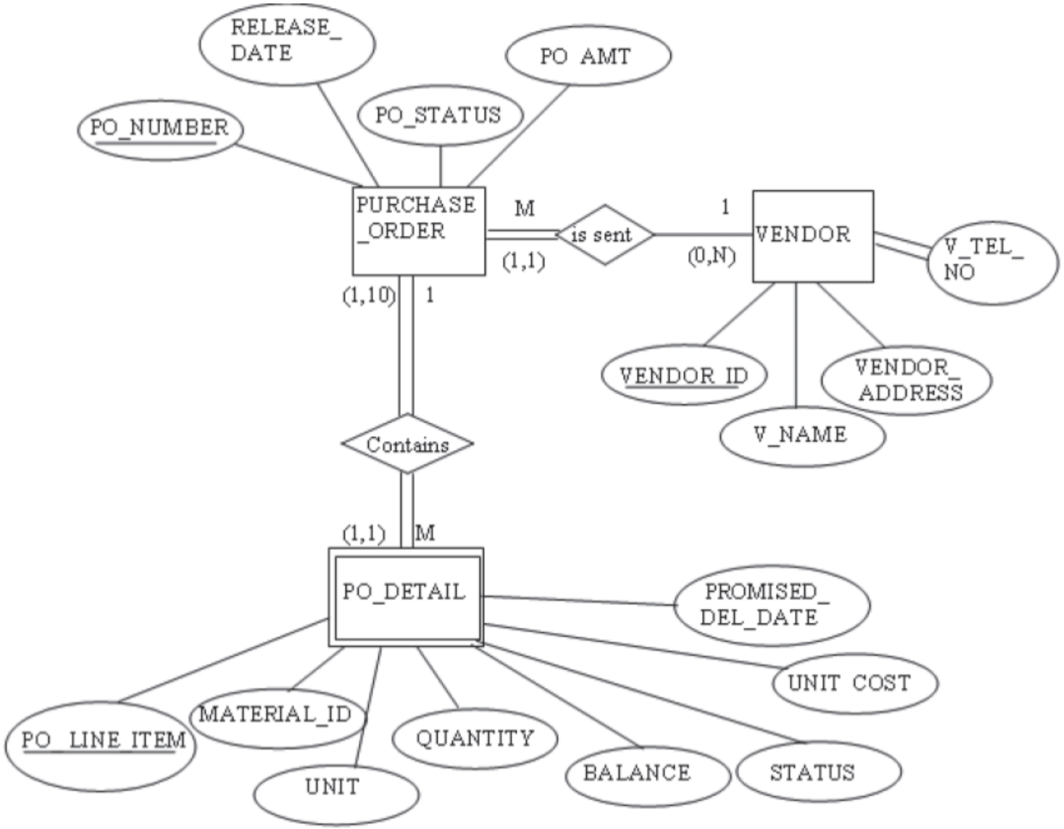
1. Optional (Partial) Participation
    1. One entity's occurrence does not require the participation of the other entity
    2. Shown by placing a single line arc near the optional entity
2. Mandatory Participation
    1. Entity requires participation of the other entity in the relationship
    2. Shown by placing a double-lined arc next to the mandatory entity

Figure 3.6 Optional participation.

a PROJECT requires the participation of an EMPLOYEE, who manages it. Therefore, a double-lined arc is placed next to PROJECT to indicate that the participation is mandatory in the relationship

2. Entity-Relationship (E-R) Modeling

2.1 E-R Modeling Primitives



2. Entity-Relationship (E-R) Modeling

2.2 The Degree of a Relationship

- The degree of a relationship is a measure of the number of entities sharing the same association.
- Four cases:
  1. Binary relationships most common
  2. Ternary relationships
  3. N-ary relationships
  4. Unary relationships

2. Entity-Relationship (E-R) Modeling

2.2 The Degree of a Relationship

- The binary relationship is most common
  - Exists when two entities have an associated relationship
  - Examples:
    - VENDOR and PURCHASE_ORDER share a binary relationship
    - PURCHASE_ORDER and PO_DETAIL share a binary relationship

2. Entity-Relationship (E-R) Modeling

2.2 The Degree of a Relationship

- The ternary relationship exists when three entities share a common relationship
  - Example: VENDOR, QUALITY_TEST, and MATERIAL entities associated with quality control testing

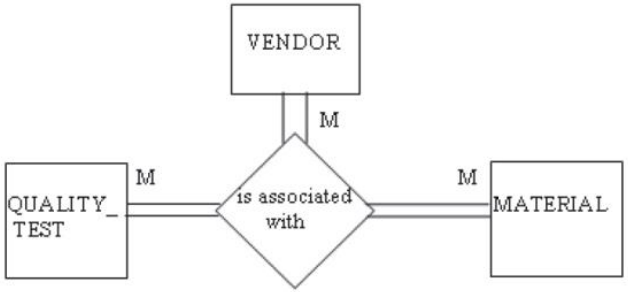
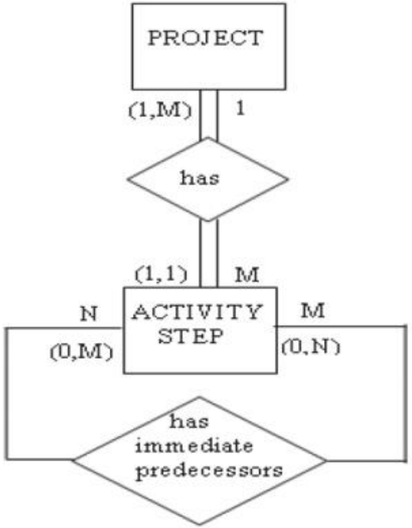


Figure 3.7 Example of a ternary relationship.

2. Entity-Relationship (E-R) Modeling

2.2 The Degree of a Relationship

- The n-ary relationship exists when more than three entities share a relationship
- The unary relationship exists when an association is maintained within a single entity (i.e., the entity has a relationship with itself)
  - This is reflected in the unary relationship of ACTIVITY_STEP with itself, which is also called a recursive relationship

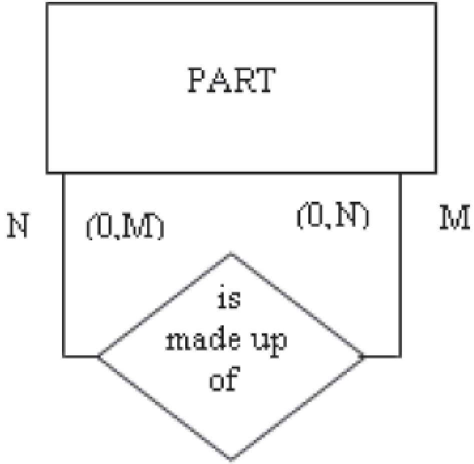


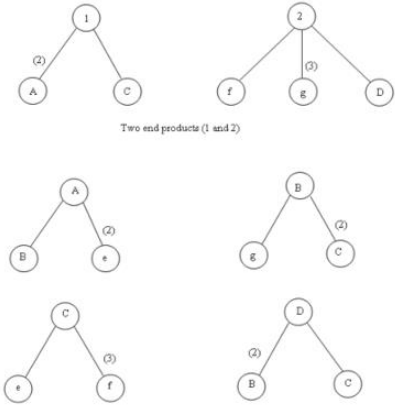
gure 3.8 Example of a unary relationship.

2. Entity-Relationship (E-R) Modeling

2.3 Recursive Entities

- Recursive entities are entities with unary relationships
- Naturally occurring in some important manufacturing problems
- Typical example: Product bill of materials (BOM), where parts are made of other parts





- Example: PART entity with “is made up of” relationship to itself, showing how complex parts are composed of simpler parts.

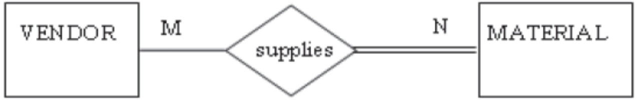
2. Entity-Relationship (E-R) Modeling

2.4 Composite Entities

- The relational model requires the use of 1:M relationships
- This enables queries to be designed that use the foreign key of the table on the M side of the relationship to retrieve corresponding unique rows on the 1 side of the relationship
- When M:N relationships are encountered in E-R diagram development, the analyst can break them down into a series of 1:M relationships using a composite entity

2. Entity-Relationship (E-R) Modeling

2.4 Composite Entities



Entity Set: VENDOR

VENDOR ID	V NAME	V STREET	V CITY	V STATE	V ZIP
V110	Jersey	2 Main St.	Patterson	NJ	07055
V25	General	125 Common	Boise	ID	44830
V250	Spices	25 Salty Lane	East Hampton	NY	10027
V75	Pasta Supply,	34 Henry St.	Philadelphia	PA	09098

Entity Set: MATERIAL

MATERIAL ID	MATL DESCRIPTION
RM201	Carrots, whole
RM202	Carrots, diced, 1/4 inch
RM205	Potatoes, Eastern,
RM210	Peas, shelled
RM211	Tomatoes, whole
RM310	Garlic, whole
RM311	Garlic powder
RM318	Salt, iodized
RM308	Onion salt
RM305	Paprika
RM340	Sugar, bulk
RM805	Olive oil
RM810	Vinegar, white

Original M:N relationship: VENDOR supplies MATERIAL

- Many vendors can supply the same material
- One vendor can supply many materials
- Creates a many-to-many relationship that needs to be resolved

2. Entity-Relationship (E-R) Modeling

2.4 Composite Entities

Entity Set: VENDOR

VENDOR ID	V NAME	V STREET	V CITY	V STATE	V ZIP
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RM305	Paprika
RM340	Sugar, bulk
RM805	Olive oil
RM810	Vinegar, white

Entity Set: VENDOR_MATL_XREF

VENDOR ID	MATERIAL ID
V25	RM201
V25	RM205
V25	RM318
V25	RM340
V110	RM201
V110	RM202
V110	RM205
V110	RM210
V110	RM211
V250	RM310
V250	RM311
V250	RM318
V250	RM340
V250	RM308
V250	RM305
V25	RM805
V25	RM810

Solution: Create a composite entity (VENDOR_MATERIAL_XREF) that breaks the M:N into two 1:M relationships

2. Entity-Relationship (E-R) Modeling

2.5 Superclass and Subclass Entity Types

- To simplify the data model while maintaining useful information:
  - Aggregate people, places, and things into as few entities as possible
  - Consistent with the semantic requirements of the problem
- Example: Raw materials, subassemblies, final assemblies, office materials, repair and maintenance materials could all be part of an entity called MATERIAL
  All instances of the MATERIAL entity have in common a material number, a description, a unit of measure (gallons, pounds), and so on
- There are attributes of MATERIAL that are unique to subgroups

2. Entity-Relationship (E-R) Modeling

2.5 Superclass and Subclass Entity Types

- A Superclass is an entity type that represents a set of similar people, places, or things
  - Have common attributes but represent distinct subclasses
- A Subclass is an entity that is a member of a superclass
  - Has distinct attributes that are not in common with all other members of the superclass

2. Entity-Relationship (E-R) Modeling

Key elements:

- Superclass Entity - main entity (e.g., EMPLOYEE)
- Specialization circle - connects superclass to subclasses
- Disjoint (d) - when every member of superclass must be in exactly one subclass
- Overlapping (o) - when members can belong to multiple subclasses
- Example: EMPLOYEE superclass with MANAGER, SALES_PERSONNEL, MACHINIST subclasses and FULL_TIME, PART_TIME subclasses